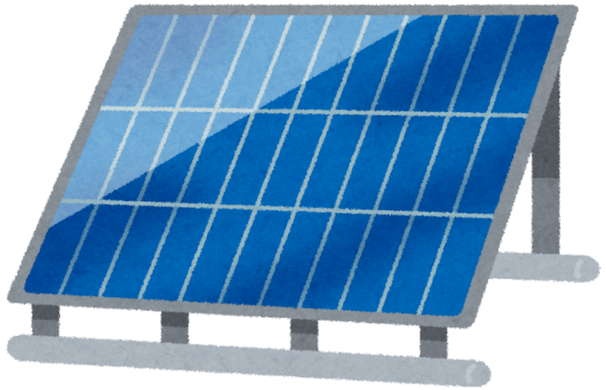


From Raw Solar Data to Actionable Insights



Solar Power Generation Analytics

DS2202 – Data Integration and Warehousing Project

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Project Scenario: Solar Power Generation Analytics

- The project analyzes solar power generation data.
- Uses three structured datasets: irradiation, weather conditions, and energy output.
- Goal: understand patterns in solar energy production.
- Approach: integrate heterogeneous datasets and study relationships between them.

Data Sources &Attribute Profiling

1

Solar Radiation and Meteorological Dataset

Type: Structured Data
Format: CSV
Records: 35,040
Notes: The dataset 2019.csv was obtained from the Solar Radiation and Meteorologica l Dataset on Kaggle.

2

Solar Power Generation – Weather Sensor Data

Type: Structured Data
Format: CSV
Records: 3,182
Notes: The Plant_1_Weather_Sensor_Data file was obtained from the Solar Power Generation dataset on Kaggle.

3

Solar Power Generation – Generation Sensor Data

Type: Structured Data
Format: CSV
Records: 68,778
Notes: The Plant_1_Generation_Data file was obtained from the Solar Power Generation dataset on Kaggle.

Initial data profiling using Kaggle tools revealed duplicate entries in some datasets, while no null values were present in any source.

For more details, refer to Lab 1 instructions:

https://www.canva.com/design/DAHKr1gsBvE/bqFPa1WXd3XIWswSkqDvdw/edit

Structured vs. Unstructured Data

Structured Data

Our project is based on three structured CSV datasets:

- Plant_1_Generation_Data
- Plant_1_Weather_Sensor_Data
- Solar Radiation & Meteorological Dataset

Why they are structured:

- Organized tables with rows and columns
- Fixed schema and clear data types

Unstructured Data

“fault_report_text"

A simulated text dataset representing technician fault descriptions.

Transformation Steps:

- Text Cleaning → Clean description
- Keyword Extraction → Fault_Type
- Regex Pattern Matching → Power_Drop_Percentage
- Location Extraction → Fault_Location
- Date Extraction → Event_Time
- Validation → Final structured fault table

Final Output Fields:

Fault_ID – PLANT_ID – DATE_TIME – Fault_Type Power_Drop_Percentage – Fault_Location -Event_Time

This unstructured source was created only for Lab 2 and not used in the final Data Warehouse.

For more details, refer to Lab 2 instructions:

https://canva.link/0rqo0tgy4cqqo1x

Data Cleaning & Preprocessing

1 Solar Radiation & Meteorological Dataset

Cleaning StepsPerformed

- *Removed the Null column (Unnamed:18)*
- *Created a sequential PK_ID as a primary key*
- *Combined Year–Month–Day–Hour–Minute into one datetime column*
- *Converted datetime to proper timestamp format*
- *Dropped the original date/time columns after merging*
- *Reordered columns to place PK_ID first*
- *Saved the cleaned dataset as a new CSV*

2 Plant_1_Generation_Data

Cleaning StepsPerformed

- *Created a sequential PK_ID as a unique primary key*
- *Reordered columns to place PK_ID as the first column*
- *Saved the cleaned dataset in CSV format for DW loading*

3 Plant_1_Weather_Sensor_Data

Cleaning StepsPerformed

- *Created a sequential PK_ID as a unique primary key*
- *Reordered columns to place PK_ID as the first column*
- *Saved the cleaned dataset in CSV format for DW loading*

Tools Used

- Python (Pandas)
- Google Colab

For more details, refer to Lab 3 instructions:

https://canva.link/rn3j7dnpdugimek

Schema Matching

How we matched the tables

We compared attributes from both datasets and identified full matches and non-matches.

Full Match

DATE_TIME = DATE_TIME
same name & meaning
PLANT_ID = PLANT_ID
identical semantics

No Match

SOURCE_KEY ≠ SOURCE_KEY
One SOURCE_KEY in the first dataset corresponds to multiple records in the second (1:n), so it is not a direct 1:1 attribute match.
AC_POWER ≠ MODULE_TEMPERATURE
Different meaning and different data type.
DAILY_YIELD ≠ AMBIENT_TEMPERATURE
One measures energy generation, the other measures temperatur

Semantic conflicts:

- Semantic conflicts occurred mainly in SOURCE_KEY and energy vs. temperature attributes.

For more details, refer to Lab 4 instructions:

https://canva.link/4ccxpspfstm7a8d

From Mapping to Data Warehouse Integration

Valid Mapping Attributes

Plant_1_Generation_Data.PLANT_ID = Plant_1_Weather_Sensor_Data.PLANT_ID
Both datasets contained the same plant identifier value (4135001).

Plant_1_Generation_Data.DATE_TIME= Plant_1_Weather_Sensor_Data.DATE_TIME
Both datasets shared the same measurement times.

Semantic Conflicts

Plant_1_Generation_Data.SOURCE_KEY ≠ Plant_1_Weather_Sensor_Data.SOURCE_KEY

SOURCE_KEY was not used for integration because:
In Generation Data → represents an inverter identifier
In Weather Data → represents a weather sensor identifier

From Mapping to Transformation :

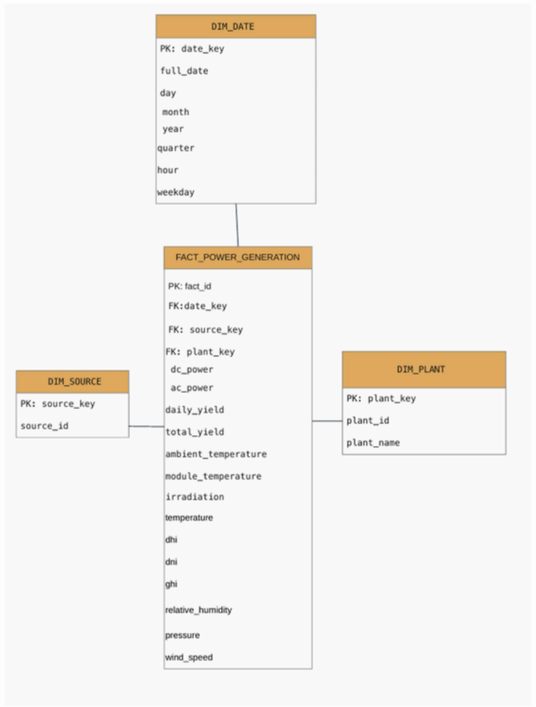
- **Merged datasets using PLANT_ID and DATE_TIME**
- **Excluded SOURCE_KEY from the merge process**

For more details, refer to Lab 5 instructions:

https://canva.link/4ccxpspfstm7a8d

Data Warehouse

Star Schema



Data Warehouse

1. DIM_DATE
- Stores date and time information.
- Key columns: date_key, full_date, day, month, year, quarter, hour, weekday.
- Used for time-based analysis and reporting.

2. DIM_PLANT
- Stores information about the solar power plant.
- Key columns: plant_key, PLANT_ID, plant_name.
- Used to identify and analyze data by plant.

3. DIM_SOURCE
- Stores inverter/source information.
- Key columns: source_key, SOURCE_ID.
- Used to analyze power generation by inverter.

4. FACT_POWER_GENERATION
- Central fact table containing power generation and weather measurements.
- Foreign Keys: **date_key**, **source_key**, **plant_key**.
- Used for analytical queries and dashboard reporting.

ETL Process

Extract
- Extracted data from three datasets:
  - Solar Power Generation Data
  - Weather Sensor Data
  - External Weather Data (2019)

Transform
- Cleaned missing and invalid values.
- Converted date and time fields into a unified format.
- Created date_key, plant_key, and source_key.
- Built the dimension tables (DIM_DATE, DIM_PLANT, and DIM_SOURCE).
- Merged datasets using PLANT_ID and DATE_TIME.
- Removed duplicate records.

Load
- Loaded the transformed data into the Data Warehouse.
- Populated the dimension tables and the FACT_POWER_GENERATION table.

Click here to view the Data Warehouse

For more details, refer to Lab 6 instructions:
https://canva.link/4ccxpspfstm7a8d

Strategic Questions & OLAP

Strategic Questions :

- What are the peak hours for solar power generation?
- Which inverter achieves the highest daily energy yield?
- How does solar irradiation affect AC power generation?
- Which weekdays achieve the highest average power generation?

OLAP Analysis Results:

Midday hours were identified as the most productive period for solar energy generation, supporting operational planning.

The best-performing inverter was identified, enabling performance evaluation and maintenance decisions.

Higher irradiation levels were associated with increased AC power generation, supporting production forecasting.

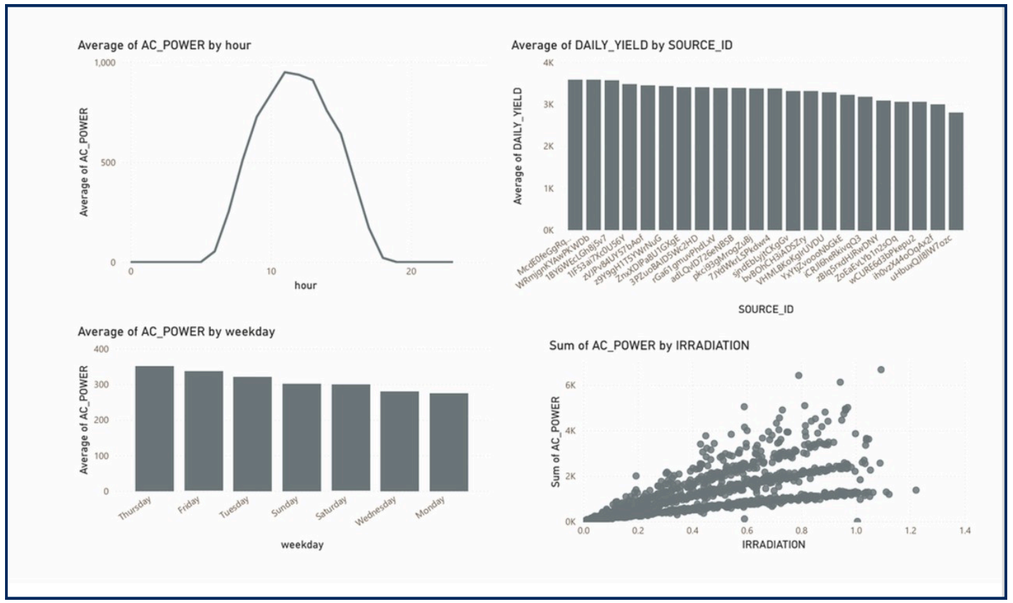
Weekly generation patterns were identified, helping monitor operational performance over time.

For more details, refer to Lab 7 instructions:

https://canva.link/4ccxpspfstm7a8d

Click here to view the Strategic Questions

Dashboards & Insights



Average AC Power by Hour

- Identified peak production periods.

Average Daily Yield by Source

- Evaluated inverter efficiency.

Average AC Power by Weekday

- Compared generation performance across weekdays.

AC Power vs Irradiation

- Analyzed environmental impact on generation.

Insights:

- Peak power generation occurs during midday hours, indicating the most productive period for solar energy generation.
- Inverter performance varies across units, highlighting opportunities for performance monitoring and maintenance.
- Higher irradiation levels are associated with increased AC power generation.
- Generation patterns differ across weekdays, indicating temporal variations in system performance.

For more details, refer to Lab 7 instructions:

https://canva.link/4ccxpspfstm7a8d

Final Recommendations &Lessons Learned

Final Recommendations



1. Integrating Fault and Maintenance Data

2. Expanding Historical Weather Data

3. Developing Predictive Models

Lessons Learned



1. Data Warehousing centralizes heterogeneous data for analytics

2.Data integration requires semantic understanding

3.Building a warehouse requires more than storing data